AQA

Please write clearly in block capitals.

Centre number | | | | | | Candidate number | | | | |

Surname _____

Forename(s) _____

Candidate signature _____

I declare this is my own work.

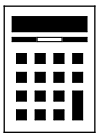
Functional Skills Level 2
MATHEMATICS

Paper 2 Calculator

Tuesday 25 February 2020 Afternoon Time allowed: 1 hour 30 minutes

Materials

For this paper you must have:



- a calculator
- mathematical instruments.

For Examiner's Use	
Question	Mark
1–5	
6	
7	
8	
9	
TOTAL	

Instructions

- Use black ink or black ball-point pen. Draw diagrams in pencil.
- Answer **all** questions.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- Do all rough work in this book. Cross through any work you do not want to be marked.
- State the units of your answer where appropriate.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 60.
- You may ask for more answer paper, graph paper and tracing paper. These must be tagged securely to this answer book.
- If your calculator does not have a π button, take the value of π to be 3.142

Advice

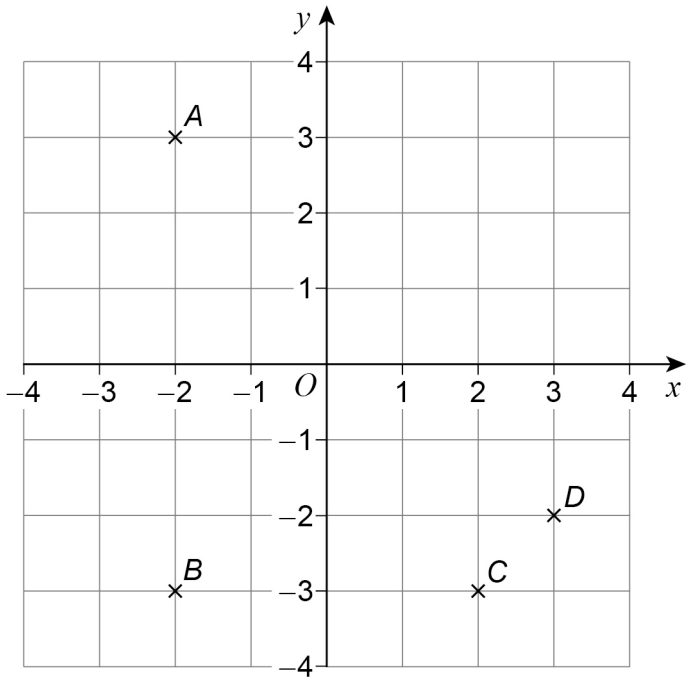
In all calculations, show clearly how you work out your answer.

Do not write outside the box

Section A

Answer **all** questions in the spaces provided.

1 Which point has the coordinates (–2, 3)?



Circle your answer.

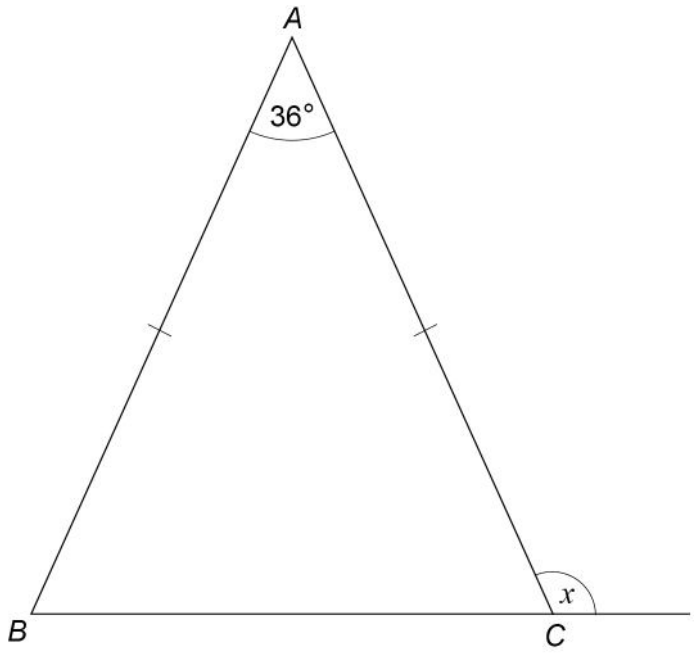
[1 mark]

A B C D

Do not write outside the box

2 Here is an isosceles triangle.

$AB = AC$



Work out the size of angle x.

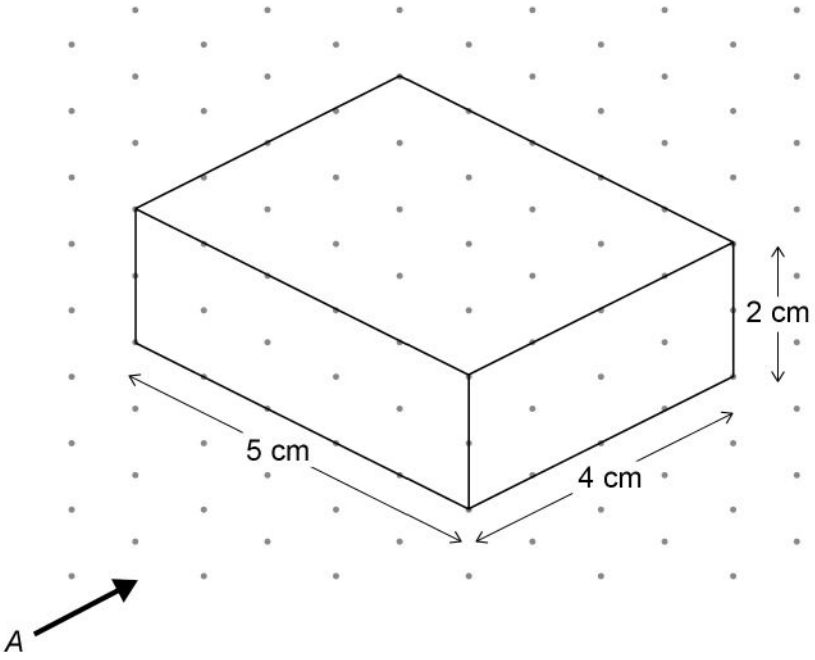
[3 marks]

Answer _____ °

Turn over ►

Do not write outside the box

3 A cuboid is shown below.



3 (a) Draw the elevation of the cuboid in the direction of the arrow.

[1 mark]

Do not write outside the box

3 (b) Work out the total surface area of the cuboid.

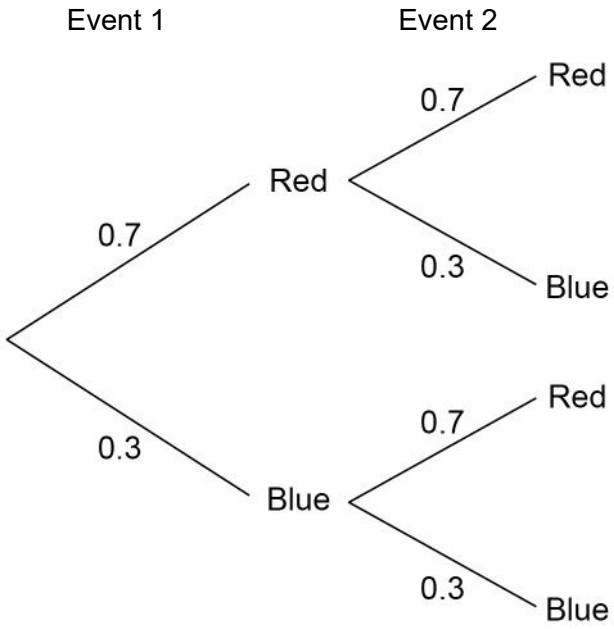
[3 marks]

Answer _____ cm^2

Turn over for the next question

Do not write outside the box

4 Here is a tree diagram.



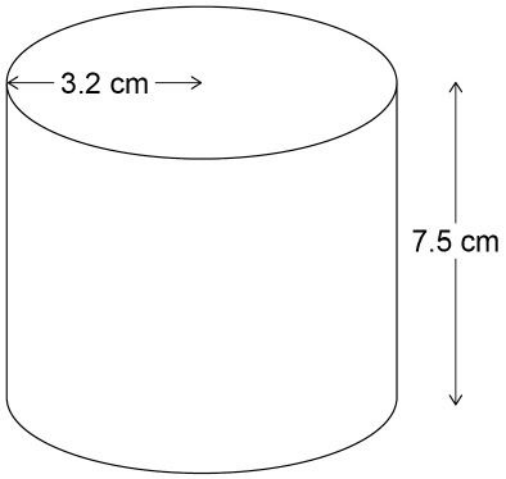
Work out the probability of Red in Event 1 **and** Red in Event 2

[2 marks]

Answer _____

Do not write outside the box

5 A cylinder has radius 3.2 cm and height 7.5 cm



Work out the volume of the cylinder.

[2 marks]

Answer _____ cm^3

12

Turn over for Section B

Do not write outside the box

Section B

Answer **all** questions in the spaces provided.

6 **Cycle race**

Paul and Fran want to enter a 60 km cycle race.

6 (a) They look at the reviews of two races, A and B, from last year.
Each review gives a score out of 5 stars to show how good the race was.
Here is some information about the review scores for **Race A**.

Mean	4.1 stars
Median	4 stars
Range	3 stars

The frequency table shows the review scores of **Race B**.

Review Score (stars)	Frequency
1	0
2	1
3	1
4	4
5	6

They enter the race with the higher average review score.

Which race do they enter?
You **must** show your working.

[4 marks]

Do not write outside the box

6 (b) Fran wants to buy a new bike for the race.

She sees this offer.

Racing bike

£708 including VAT at 20%

SPECIAL OFFER

WE PAY THE VAT

Fran has £600 to spend.

Can she afford the bike?
You **must** show your working.

[3 marks]

Question 6 continues on the next page

Do not write outside the box

6 (c) Paul and Fran start the race at the same time.

Fran cycles the 60 km at an average speed of 15 **miles** per hour.

Fran completes the race and waits for Paul to finish.

Paul finishes the race in 167 minutes.

Does Fran wait for **more than** 10 minutes?
Use 1 mile = 1.6 km
You **must** show your working.

[4 marks]

11

Do not write outside the box

7 Supermarket

Lucas is going shopping at a supermarket.

7 (a) Lucas has a budget of £50 per week for food.
He has already spent £12.50 on takeaway meals.

What fraction of his budget does he have left?

[2 marks]

Answer _____

Question 7 continues on the next page

Do not write outside the box

7 (b) Lucas buys a 450 g packet of biscuits.



He reads the label.

Sugar content	26.4 g per 100 g
Maximum recommended daily amount (MRDA) of sugar	30 g

Lucas eats 2 biscuits.

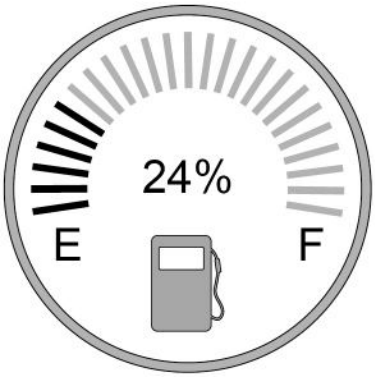
Work out the percentage of the MRDA of sugar in the 2 biscuits.

[4 marks]

Answer _____ %

Do not write outside the box

7 (c) Lucas looks at his car's petrol gauge and decides to fill up the petrol tank.



The petrol tank holds 70 litres when full.

Petrol costs 126.8p per litre.

How much does Lucas pay to fill up the tank?

[4 marks]

Answer £ _____

10

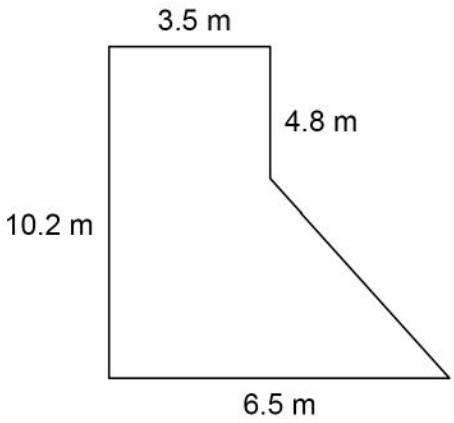
Turn over ►

Do not write outside the box

8 **Driveway**

Meg owns a company that builds concrete driveways.

8 (a) Here is a sketch of a plan of a driveway made from a rectangle and a triangle.



Not drawn accurately

Meg charges £120 per square metre to construct the driveway.

Work out the total amount she charges for the driveway.

[4 marks]

Answer £ _____

Do not write outside the box

8 (b) Meg promises the customer that the driveway will be constructed in 7 days.

Meg has 3 workers.

It would take the 3 workers 11 days to construct the driveway.

Work out the minimum number of **extra** workers Meg needs to employ.
Assume all workers work at the same rate.
You **must** show your working.

[4 marks]

Answer _____

Question 8 continues on the next page

Do not write outside the box

8 (c) Meg paints a different driveway with a waterproof coating.

The driveway covers an area of 35.75 m^2

The coating is made by mixing sealant and water in the ratio 3 : 8

2 litres of the coating will cover 1 m^2 of driveway.

Sealant is sold in 1.5 litre bottles.

How many bottles of sealant does she need?

[5 marks]

Answer _____

13

Do not write outside the box

9 Apartment

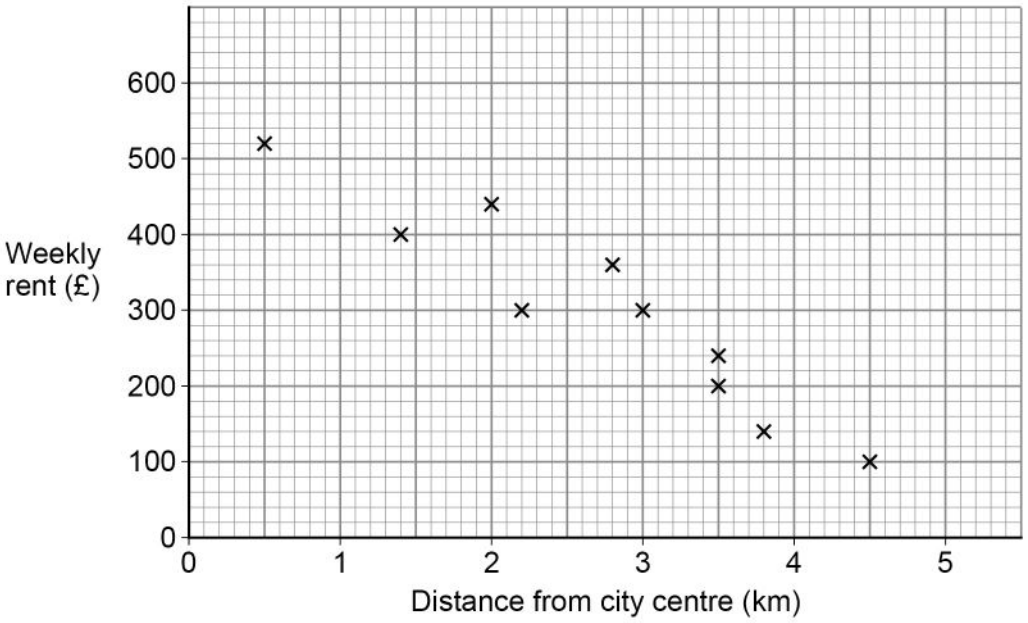
Pete wants to rent an apartment near Norwich city centre.

9 (a) He records information about 10 apartments on a scatter diagram.

The diagram shows

the distance from the city centre in km

the weekly rent in £



The table shows extra data about two other apartments.

Distance from city centre (km)	Weekly rent (£)
2	250
3.2	180

Pete rents an apartment for £280 per week.

Use the scatter diagram **with the extra data** to

estimate the distance from the city centre to the apartment.

You **must** show your working, which should be on the diagram.

[4 marks]

Answer _____

Question 9 continues on the next page

Do not write outside the box

9 (b) Pete has a scale diagram of the living room in the apartment.

Pete wants to cover one wall with wallpaper.

The width of the wall on the scale drawing is 15.9 cm

The scale is 1 : 60

The wallpaper Pete wants

comes in rolls of width 0.53 m and length 10.05 m

costs £13.85 per roll.

The height of the room is 2.65 m

Pete cuts the rolls into lengths of 2.65 m

Work out the total cost of the rolls of wallpaper that Pete needs for the wall.

[6 marks]

Answer £ _____

Do not write outside the box

9 (c) Pete wants to hide the wires from his TV.

Wires can be hidden in plastic trunking.

Trunking can be cut to any length.

Pete needs three pieces of trunking measuring

$1\frac{1}{2}$ metres

$2\frac{1}{4}$ metres

90 cm

Pete has 15 feet of trunking.

Is this enough?

Use 1 foot = 12 inches

Use 1 inch = 2.5 cm

[4 marks]

14

END OF QUESTIONS

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ANSWER IN THE SPACES PROVIDED

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Question number	**Additional page, if required.** **Write the question numbers in the left-hand margin.**

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